Shree Harsha Bokkahalli Satish

PhD Student in Speech Technology and Conversational AI

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Research Profile

Doctoral student specialising in speech technology and conversational AI. Current research focus on distributional evaluation methods for full-duplex dialogue and examining how conversational behaviours and socially meaningful vocal cues affect the performance, robustness, and fairness of speech foundation models.

Education

KTH Royal Institute of Technology **2025–2029 (expected)**
PhD in speech technology and conversational AI
Doctoral student in the Wallenberg AI, Autonomous Systems and Software Program (WASP)

University of Liverpool **2022–2024**
MSc Data Science and Artificial Intelligence, Distinction
Year in industry developing LLM/RAG systems for aerospace engineering workflows.

Indian Institute of Technology Bombay **2018–2021**
MTech Electrical Engineering (Signal Processing), GPA: 8.18/10
TATA Fellow; research in speech technology for social impact.

Sri Jayachamarajendra College of Engineering, Mysuru **2014–2018**
BE Electronics and Communication Engineering, GPA: 8.5/10
Graduate Aptitude Test in Engineering rank: 749th nationally (99th percentile).

Research and Professional Experience

Doctoral Researcher **Mar 2025–Present**
KTH Royal Institute of Technology, Department of Speech, Music and Hearing (TMH) *Stockholm, Sweden*

- Developed human-reference, distributional metrics for comparing human–human, human–AI, and AI–AI conversations, with emphasis on turn-taking, responsiveness, entrainment, affect, and listener judgments.
- Designed controlled evaluations of SpeechLLMs across accent, gender presentation, phonation, speech continuation, positional bias, benchmark format, and open-ended generation tasks.
- Studied audio deepfake detection under speech restoration and voice-quality conversion, analysing distribution shifts and robustness to benign transformations.

Innovation Engineer, Year in Industry **Aug 2023–Sep 2024**
ALTEN Ltd. *Derby, UK*

- Built a multi-agent retrieval-augmented generation system for aerospace repair-scheme creation; internal evaluation indicated a 98% reduction in engineering workflow time.
- Delivered a near-production prototype through iterative development and continuous feedback from engineering stakeholders.

Postgraduate Research Assistant and TATA Fellow **Jun 2018–Aug 2021**
Digital Audio Processing Lab, IIT Bombay *Mumbai, India*

- Developed speech recognition (acoustic and language models) and automated literacy-assessment systems for children's oral reading; a partner NGO deployed the resulting system in field programmes.
- Improved robustness under challenging field conditions using acoustic modelling, data augmentation, transfer learning, and voice activity detection.

Skills

Methods: Speech foundation models, SpeechLLMs, speech continuation, full-duplex dialogue, distributional evaluation, audio deepfake detection, self-supervised speech representations, ASR, bias and fairness evaluation, reward modelling.
Tools: Python, PyTorch, scikit-learn, MATLAB, Bash, Kaldi, and LLM/RAG systems.

Peer-Reviewed Publications

[1] **The Voice Behind the Words: Quantifying Intersectional Bias in SpeechLLMs**
Shree Harsha Bokkahalli Satish, Christoph Minixhofer, Maria Teleki, James Caverlee, Ondřej Klejch, Peter Bell, Gustav Eje Henter, and Éva Székely
Proceedings of Interspeech 2026

[2] **Lost in Phonation: Voice Quality Variation as an Evaluation Dimension for Speech Foundation Models**
Harm Lameris, **Shree Harsha Bokkahalli Satish**, Joakim Gustafson, and Éva Székely
Proceedings of Interspeech 2026 (accepted)

[3] **From Seeing it to Experiencing it: Interactive Evaluation of Intersectional Voice Bias in Human-AI Speech Interaction**
Shree Harsha Bokkahalli Satish, Maria Teleki, Christoph Minixhofer, Ondrej Klejch, Peter Bell, and Éva Székely
CHI 2026 Extended Abstracts

[4] **"Walk a Mile in My Voice": Voice Conversion Shapes Trust, Attribution, and Empathy in Human-AI Speech Interactions**
Shree Harsha Bokkahalli Satish, Maria Teleki, Christoph Minixhofer, Ondrej Klejch, Peter Bell, and Éva Székely
IUI 2026 Companion, pp. 126–130

[5] **Do Bias Benchmarks Generalise? Evidence from Voice-based Evaluation of Gender Bias in SpeechLLMs**
Shree Harsha Bokkahalli Satish, Gustav Eje Henter, and Éva Székely
IEEE ICASSP 2026 (accepted)

[6] **Speak Your Mind: The Speech Continuation Task as a Probe of Voice-Based Model Bias**
Shree Harsha Bokkahalli Satish, Harm Lameris, Olivier Perrotin, Gustav Eje Henter, and Éva Székely
Identity-Aware AI Workshop at LREC 2026

[7] **When Voice Matters: Evidence of Gender Disparity in Positional Bias of SpeechLLMs**
Shree Harsha Bokkahalli Satish, Gustav Eje Henter, and Éva Székely
Speech and Computer (SPECOM 2025), pp. 25–38

[8] **Hear Me Out: Interactive Evaluation and Bias Discovery Platform for Speech-to-Speech Conversational AI**
Shree Harsha Bokkahalli Satish, Gustav Eje Henter, and Éva Székely
Proceedings of Interspeech 2025, pp. 2151–2152

Preprints and Accepted Presentations

[P1] **Distributional Metrics for Evaluating Spoken Conversational Systems**
Shree Harsha Bokkahalli Satish, Erica Cooper, Patrícia Schmidtová, Maike Züfle, Éva Székely, Nicholas Sanders, and Ondřej Klejch
arXiv:2609.31719, 2026; submitted to IEEE ICASSP 2027

[P2] **A Cross-Community Agenda for Speech AI**
Maria Teleki*, Kimi V. Wenzel*, Anna Seo Gyeong Choi*, Tobias Weinberg*, **Shree Harsha Bokkahalli Satish***, Stephanny Sanchez*, Belu Ticona*, Ariadna Sanchez*, Yash Sonkar*, Aarti Mathur*, Christoph Minixhofer*, Abraham Glasser*, Raja Kushalnagar*, James Caverlee*, Minha Lee*, Shaomei Wu*, Alyssa Hillary Zisk*, Éva Székely*, Dylan Gaines*, Angelika Seeschaaf Veres*, Seray Ibrahim*, Nicholas Cummins*, and Allison Koenecke*
Submitted to ACL Rolling Review (ARR) 2026

[P3] **What Counts as Real? Speech Restoration and Voice Quality Conversion Pose New Challenges to Deepfake Detection**
Shree Harsha Bokkahalli Satish, Harm Lameris, Joakim Gustafson, and Éva Székely
Accepted for presentation at the International Symposium on Synthetic Media Attribution and Detection (ISSMAD 2026); arXiv:2603.14033

[P4] **Predicting Lexical Skills from Oral Reading with Acoustic Measures**
Charvi Vitthal*, **Shreeharsha B. S***, Kamini Sabu, and Preeti Rao
arXiv:2112.00635, 2021

*Equal contribution.

Selected Awards and Achievements

- Invited participant to the 2026 JSALT workshop at the Johns Hopkins university, working on simulation and evaluation for full-duplex conversational models.
- EnAppSys/Montel Datathon: 2nd place on the public leaderboard and 3rd on the private leaderboard for energy forecasting.
- IIT Madras English/Hindi ASR Challenge: 2nd place in the closed task.
- Interspeech 2020 Shared Task on ASR: 9th place in the closed task; achieved an 8.5% relative WER improvement using wavelet based data augmentation.